

ADVANCED CYBERSECURITY PRACTICAL ASSESSMENT

WEEK 3 – NETWORK SECURITY & VULNERABILITY ASSESSMENT

Internship Domain: Cybersecurity

Submitted By:

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Role: Cybersecurity Intern

Tools Used:

- Kali Linux
- Ubuntu 24.04.5 LTS
- Nmap
- Wireshark
- UTM

Target IP: 192.168.64.3

1. INTRODUCTION

Cybersecurity is the practice of protecting computer systems, networks, devices, and data from unauthorized access, attacks, and security threats. Network security assessment is an important part of cybersecurity because it helps identify active hosts, open ports, running services, and possible security weaknesses.

This Week 3 practical assessment focuses on network discovery, Nmap scanning, Wireshark traffic analysis, vulnerability assessment, and security hardening. The practical work was performed in an authorized laboratory environment using two virtual machines: Kali Linux as the security-testing system and Ubuntu 24.04.5 LTS as the target system. During the assessment, the Ubuntu target was identified on the laboratory network with the IP address 192.168.64.3. Nmap was used to discover the target and examine its network services. The assessment identified SSH on TCP port 22 and detected the running service as OpenSSH 9.6p1 on Ubuntu.

Wireshark was also used to capture and analyze network traffic. The captured packets help understand communication between the Kali Linux testing machine and the Ubuntu target. Different protocols and packet types can be examined to understand how network communication takes place.

The main objective of this assessment is to gain practical experience in identifying network services, analyzing traffic, recognizing potential security issues, and recommending appropriate security-hardening measures. All testing was performed only against the authorized virtual machines created for this cybersecurity laboratory.

Objectives

- To identify active hosts on the laboratory network.
- To perform TCP and UDP port scanning using Nmap.
- To identify running services and their versions.
- To perform basic operating-system detection.
- To capture and analyze network traffic using Wireshark.
- To investigate potential security weaknesses.
- To recommend suitable security-hardening measures.
- To document the complete assessment with screenshots and evidence.

Tools Used: Kali Linux, Nmap, Wireshark, Ubuntu 24.04.5 LTS and UTM.

LAB ENVIRONMENT AND NETWORK TOPOLOGY

The practical assessment was performed in a controlled virtual laboratory environment using UTM on an Apple Silicon Mac. Two virtual machines were used for the assessment.

Lab Machines

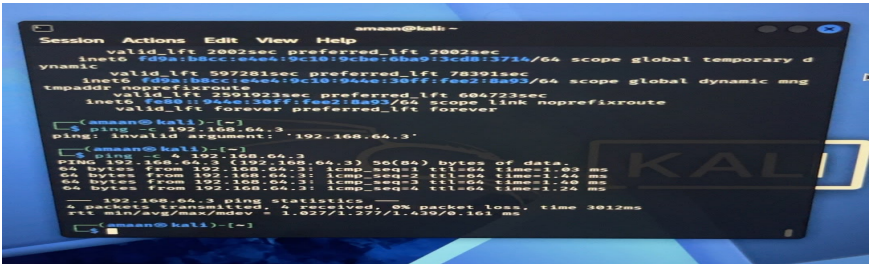
Machine	Purpose	IP Address
Kali Linux	Security testing and network analysis	192.168.64.2
Ubuntu 24.04.5 LTS	Target machine	192.168.64.3

Tools Used

- Kali Linux – Security testing platform
- Nmap 7.99 – Network discovery and port/service scanning
- Wireshark – Network packet capture and traffic analysis
- Ubuntu 24.04.5 LTS – Target system
- UTM – Virtualization platform

Network Configuration

Both virtual machines were connected to the same UTM shared network. The Kali Linux machine was assigned the IP address 192.168.64.2, while the Ubuntu target was assigned 192.168.64.3. Connectivity between the two systems was verified using ICMP ping. The test from Kali to Ubuntu returned 4 packets transmitted, 4 received, with 0% packet loss, confirming that the two machines could communicate successfully.



Connectivity verification between Kali Linux and the Ubuntu target.

Assessment Approach

Network Discovery → Port Scanning → Service Detection → OS Detection → UDP Scanning → Vulnerability Assessment → Wireshark Traffic Analysis → Security Findings → Hardening Recommendations

TASK 7 – NETWORK DISCOVERY

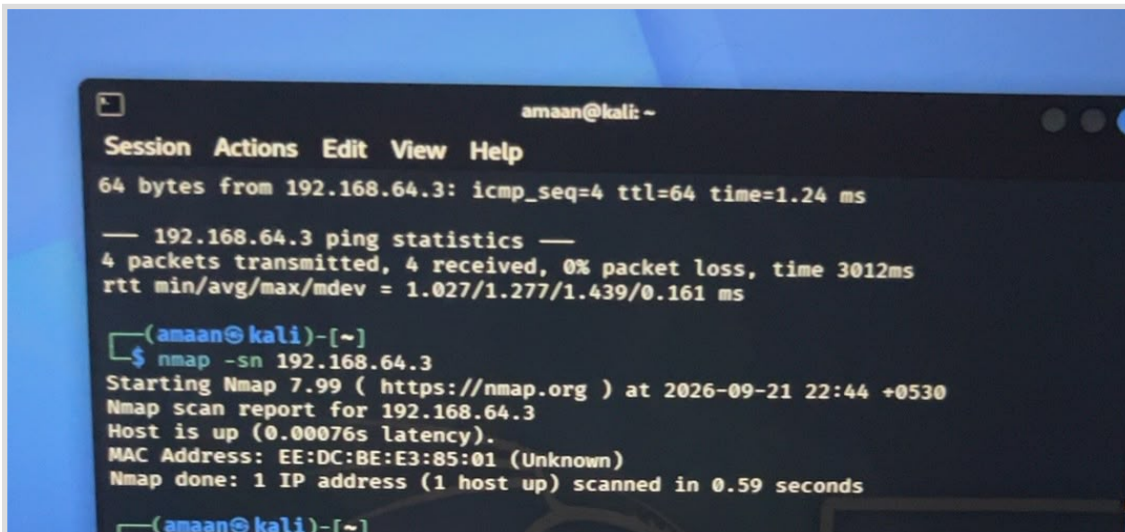
Network discovery is the first stage of a network security assessment. Its purpose is to identify active systems and determine which hosts are reachable on the laboratory network. In this assessment, Kali Linux was used as the testing machine and Ubuntu 24.04.5 LTS was used as the authorized target. The Ubuntu target was assigned to the IP address 192.168.64.3.

Host Discovery

Nmap command was used to verify whether the target was active: `nmap -sn 192.168.64.3`

Basic Port Discovery

A basic Nmap scan was then performed: `nmap 192.168.64.3`

A screenshot of a terminal window titled 'amaan@kali: ~'. The window shows the output of a ping command and an Nmap scan. The ping command 'ping -c 4 192.168.64.3' shows 64 bytes from 192.168.64.3 with icmp_seq=4, ttl=64, and time=1.24 ms. Below this, the Nmap scan report for 192.168.64.3 is displayed, showing that the host is up with a latency of 0.00076s. The scan was completed in 0.59 seconds.

```
Session Actions Edit View Help
64 bytes from 192.168.64.3: icmp_seq=4 ttl=64 time=1.24 ms

— 192.168.64.3 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3012ms
rtt min/avg/max/mdev = 1.027/1.277/1.439/0.161 ms

(amaan@kali)-[~]
$ nmap -sn 192.168.64.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 22:44 +0530
Nmap scan report for 192.168.64.3
Host is up (0.00076s latency).
MAC Address: EE:DC:BE:E3:85:01 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 0.59 seconds

(amaan@kali)-[~]
```

Nmap host discovery of the Ubuntu target

TASK 8 – ADVANCED NMAP SCANNING

Advanced Nmap scanning was performed to obtain more detailed information about the Ubuntu target. The assessment included service and version detection, operating-system detection, TCP SYN scanning, UDP scanning, and vulnerability-script scanning.

Service and Version Detection

Command was used: `nmap -sV 192.168.64.3`

The scan identified SSH on TCP port 22 and detected the service as:

OpenSSH 9.6p1 Ubuntu 3ubuntu13.19

Operating System Detection

Command was used: `sudo nmap -O 192.168.64.3`

Nmap identified the target as a Linux-based system.

TCP SYN Scan

Command was used: `sudo nmap -sS 192.168.64.3`

The scan confirmed that 22/tcp (SSH) was open

UDP Scan

Command was used: `sudo nmap -sU --top-ports 20 192.168.64.3`

The scan showed a mixture of closed and open/filtered UDP ports

Vulnerability Script Scan

Command was used: `sudo nmap --script vuln 192.168.64.3`

The scan identified the SSH service on port 22 but did not report a specific vulnerability in the displayed results.

```
amaan@kali: ~  
Session Actions Edit View Help  
Failed to resolve "sv".  
Nmap scan report for 192.168.64.3  
Host is up (0.00054s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: EE:DC:BE:E3:85:01 (Unknown)  
Nmap done: 1 IP address (1 host up) scanned in 1.21 seconds  
  
amaan@kali)~  
$ nmap -sV 192.168.64.3  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 22:55 +0530  
Nmap scan report for 192.168.64.3  
Host is up (0.00051s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.19 (Ubuntu Linux; protocol 2.0)  
MAC Address: EE:DC:BE:E3:85:01 (Unknown)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 0.89 seconds  
  
amaan@kali)~  
$
```

Nmap service and version detection.

```
amaan@kali: ~  
Session Actions Edit View Help  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 2.00 seconds  
  
amaan@kali)~  
$ sudo nmap -O 192.168.64.3  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 23:03 +0530  
Nmap scan report for 192.168.64.3  
Host is up (0.00073s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: EE:DC:BE:E3:85:01 (Unknown)  
Device type: general purpose/router  
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X  
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3  
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)  
Network Distance: 1 hop  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 2.00 seconds  
  
amaan@kali)~  
$
```

Nmap operating-system detection.

```
amaan@kali: ~  
Session Actions Edit View Help  
22/tcp    open  ssh  
MAC Address: EE:DC:BE:E3:85:01 (Unknown)  
Device type: general purpose/router  
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X  
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3  
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)  
Network Distance: 1 hop  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 2.00 seconds  
  
amaan@kali)~  
$ sudo nmap -sS 192.168.64.3  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 23:05 +0530  
Nmap scan report for 192.168.64.3  
Host is up (0.00047s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: EE:DC:BE:E3:85:01 (Unknown)  
Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds  
  
amaan@kali)~  
$
```

TCP SYN scan of the authorized Ubuntu target.

TASK 9: WIRESHARK NETWORK TRAFFIC CAPTURE

Wireshark was used to capture and analyze network traffic between the Kali Linux system and the authorized Ubuntu target. The purpose was to observe different protocols and understand how network packets are transmitted.

Starting Wireshark

Wireshark was opened in Kali Linux and the active network interface was selected. The capture was started before generating network traffic.

Wireshark Display Filters

following filters were used to examine specific types of traffic:

Icmp: Used to display ICMP packets generated by ping

Tcp: Used to display TCP packets

Udp: Used to display UDP packets

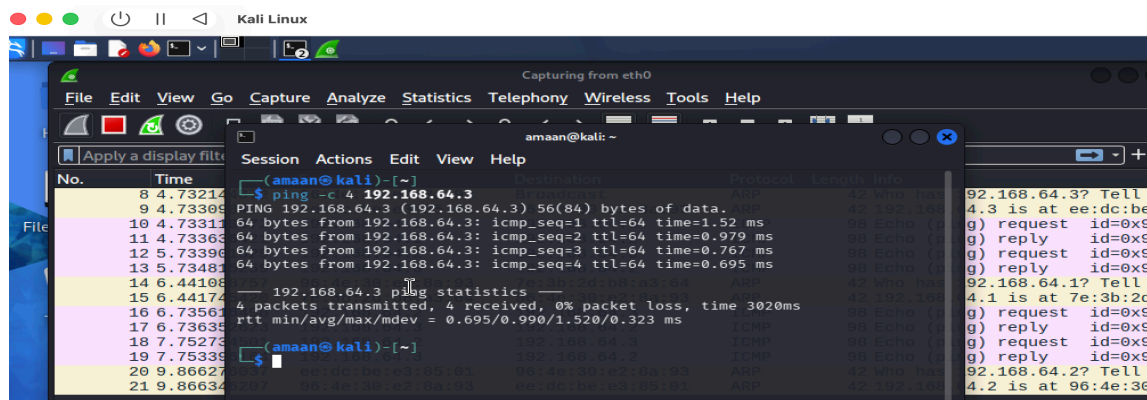
Arp: Used to display ARP packets

Dns: Used to display DNS queries and responses

ICMP Traffic Analysis

A ping test was performed from Kali to the Ubuntu target: `ping -c 4 192.168.64.3`

The test successfully received four ICMP replies with 0% packet loss. The corresponding ICMP packets could be observed in Wireshark.

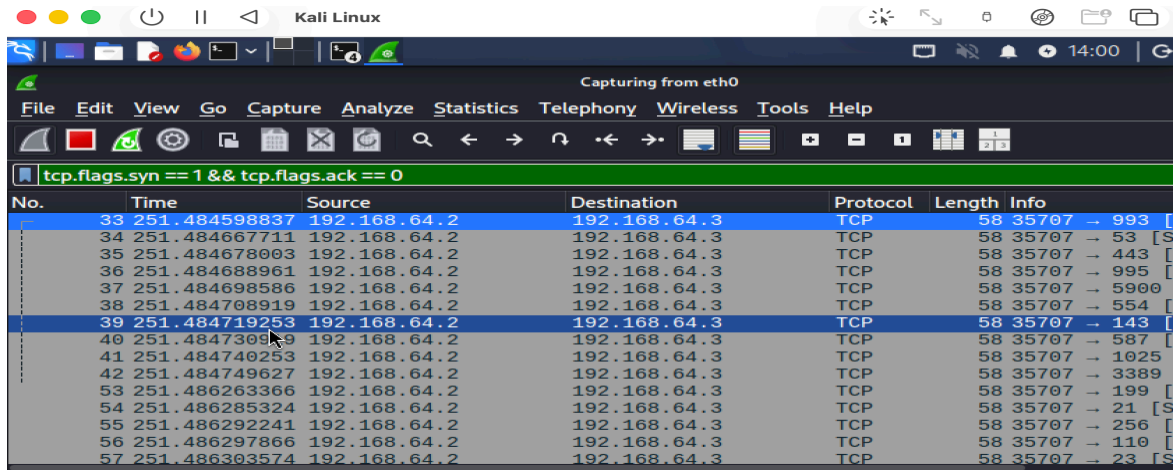


Wireshark live capture showing ICMP traffic between Kali Linux and the Ubuntu target

TASK 10: NMAP + WIRESHARK PACKET INVESTIGATION

Start Wireshark Capture: Keep Wireshark open and make sure it is capturing on eth0.

Wireshark captured TCP SYN packets generated during the Nmap SYN scan. The packets originated from the Kali machine (192.168.64.2) and were sent to the Ubuntu target (192.168.64.3). The SYN packets represent connection attempts used by Nmap to identify open TCP ports.



The image shows the Wireshark network protocol analyzer interface. The top bar indicates it is capturing on the eth0 interface. The packet list pane shows a filter applied: `tcp.flags.syn == 1 && tcp.flags.ack == 0`. The packet list contains 15 entries, all of which are TCP SYN packets. The selected packet (No. 39) has a time of 251.484719253, source IP of 192.168.64.2, and destination IP of 192.168.64.3. The packet details pane shows the TCP header with flags SYN and ACK=0.

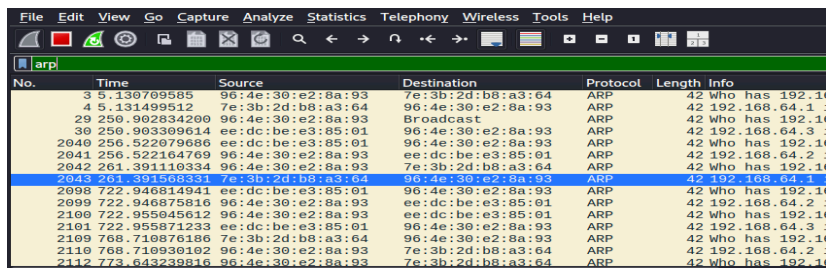
No.	Time	Source	Destination	Protocol	Length	Info
33	251.484598837	192.168.64.2	192.168.64.3	TCP	58	35707 → 993 [S
34	251.484667711	192.168.64.2	192.168.64.3	TCP	58	35707 → 53 [S
35	251.484678003	192.168.64.2	192.168.64.3	TCP	58	35707 → 443 [S
36	251.484688961	192.168.64.2	192.168.64.3	TCP	58	35707 → 995 [S
37	251.484698586	192.168.64.2	192.168.64.3	TCP	58	35707 → 5900 [S
38	251.484708919	192.168.64.2	192.168.64.3	TCP	58	35707 → 554 [S
39	251.484719253	192.168.64.2	192.168.64.3	TCP	58	35707 → 143 [S
40	251.484730919	192.168.64.2	192.168.64.3	TCP	58	35707 → 587 [S
41	251.484740253	192.168.64.2	192.168.64.3	TCP	58	35707 → 1025 [S
42	251.484749627	192.168.64.2	192.168.64.3	TCP	58	35707 → 3389 [S
53	251.486263366	192.168.64.2	192.168.64.3	TCP	58	35707 → 199 [S
54	251.486285324	192.168.64.2	192.168.64.3	TCP	58	35707 → 21 [S
55	251.486292241	192.168.64.2	192.168.64.3	TCP	58	35707 → 256 [S
56	251.486297866	192.168.64.2	192.168.64.3	TCP	58	35707 → 110 [S
57	251.486303574	192.168.64.2	192.168.64.3	TCP	58	35707 → 23 [S

Task 11: Traffic Investigation

To capture and analyze different types of network traffic between the Kali Linux system and the Ubuntu target using Wireshark

Traffic analyzed:

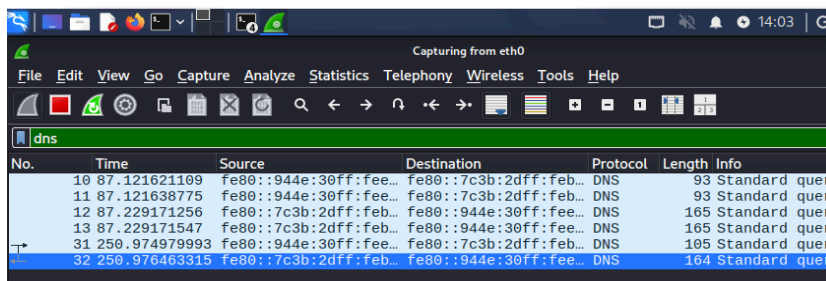
1. ARP: Used for resolving IP addresses to MAC addresses on the local network.
2. DNS: Used for domain-name resolution.
3. TCP: Used for reliable, connection-oriented communication.
4. UDP: Used for connectionless communication with lower overhead.
5. ICMP: Used for network connectivity testing, such as ping.



A screenshot of the Wireshark interface showing a list of ARP traffic. The packet list pane on the left shows several ARP packets. The main packet details pane shows the structure of an ARP request, including the Ethernet II header, ARP header, and the payload containing the source and target MAC and IP addresses.

No.	Time	Source	Destination	Protocol	Length	Info
3	5.130709585	96:4e:30:e2:8a:93	7e:3b:2d:b8:a3:64	ARP	42	who has 192.168.64.2
4	5.131499512	7e:3b:2d:b8:a3:64	96:4e:30:e2:8a:93	ARP	42	192.168.64.1
29	250.902834200	96:4e:30:e2:8a:93	Broadcast	ARP	42	who has 192.168.64.2
30	250.903309614	ee:dc:be:e3:85:01	96:4e:30:e2:8a:93	ARP	42	192.168.64.3
2040	256.522079686	ee:dc:be:e3:85:01	96:4e:30:e2:8a:93	ARP	42	who has 192.168.64.2
2041	256.522164769	96:4e:30:e2:8a:93	ee:dc:be:e3:85:01	ARP	42	192.168.64.2
2042	261.391110334	96:4e:30:e2:8a:93	7e:3b:2d:b8:a3:64	ARP	42	who has 192.168.64.2
2043	261.391508331	7e:3b:2d:b8:a3:64	96:4e:30:e2:8a:93	ARP	42	192.168.64.2
2093	722.946814941	ee:dc:be:e3:85:01	96:4e:30:e2:8a:93	ARP	42	who has 192.168.64.2
2099	722.946875816	96:4e:30:e2:8a:93	ee:dc:be:e3:85:01	ARP	42	192.168.64.2
2100	722.955045612	96:4e:30:e2:8a:93	ee:dc:be:e3:85:01	ARP	42	who has 192.168.64.2
2101	722.955871233	ee:dc:be:e3:85:01	96:4e:30:e2:8a:93	ARP	42	192.168.64.3
2109	768.710870186	7e:3b:2d:b8:a3:64	96:4e:30:e2:8a:93	ARP	42	who has 192.168.64.2
2110	768.710930192	96:4e:30:e2:8a:93	7e:3b:2d:b8:a3:64	ARP	42	192.168.64.2
2112	773.043239816	96:4e:30:e2:8a:93	7e:3b:2d:b8:a3:64	ARP	42	who has 192.168.64.2

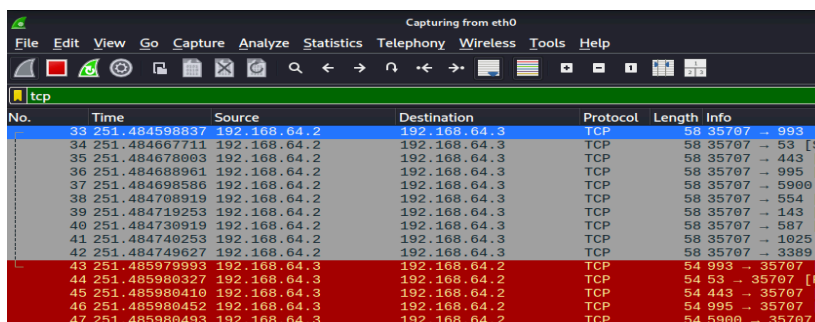
ARP Traffic Capture in Wireshark



A screenshot of the Wireshark interface showing a list of DNS traffic. The packet list pane on the left shows several DNS queries. The main packet details pane shows the structure of a DNS query, including the Ethernet II header, IP header, and the DNS query packet.

No.	Time	Source	Destination	Protocol	Length	Info
10	87.121621109	fe80::944e:30ff:feb...	fe80::7c3b:2dff:feb...	DNS	93	Standard query
11	87.121638775	fe80::944e:30ff:feb...	fe80::7c3b:2dff:feb...	DNS	93	Standard query
12	87.229171256	fe80::7c3b:2dff:feb...	fe80::944e:30ff:feb...	DNS	165	Standard query
13	87.229171547	fe80::7c3b:2dff:feb...	fe80::944e:30ff:feb...	DNS	165	Standard query
31	250.974979993	fe80::944e:30ff:feb...	fe80::7c3b:2dff:feb...	DNS	105	Standard query
32	250.976463315	fe80::7c3b:2dff:feb...	fe80::944e:30ff:feb...	DNS	164	Standard query

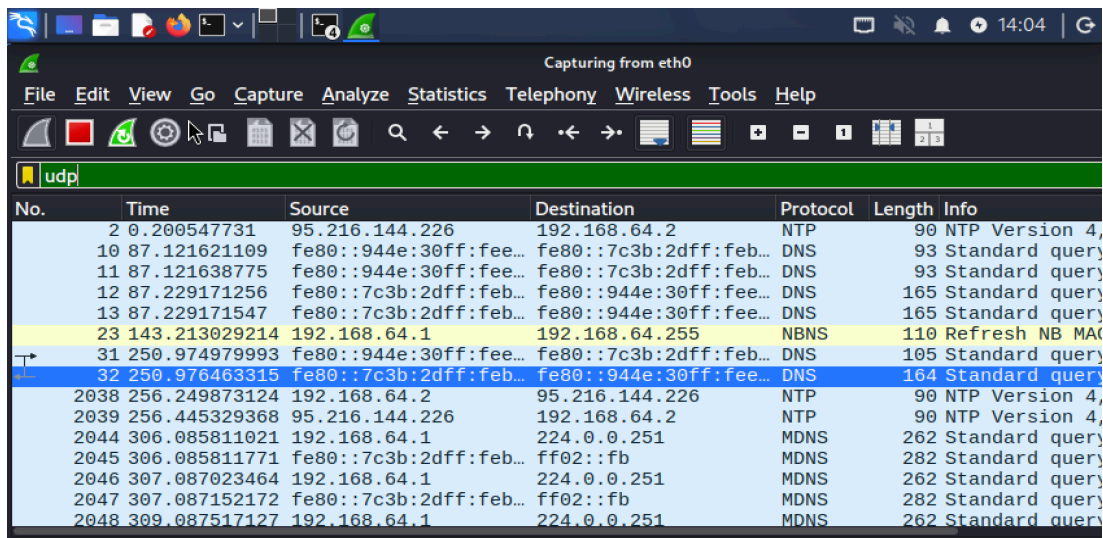
DNS Traffic Capture in Wireshark



A screenshot of the Wireshark interface showing a list of TCP traffic. The packet list pane on the left shows several TCP segments. The main packet details pane shows the structure of a TCP segment, including the Ethernet II header, IP header, and the TCP segment.

No.	Time	Source	Destination	Protocol	Length	Info
33	251.484598837	192.168.64.2	192.168.64.3	TCP	58	35707 → 993 [S]
34	251.484667711	192.168.64.2	192.168.64.3	TCP	58	35707 → 53 [S]
35	251.484678003	192.168.64.2	192.168.64.3	TCP	58	35707 → 443 [S]
36	251.484688961	192.168.64.2	192.168.64.3	TCP	58	35707 → 995 [S]
37	251.484698586	192.168.64.2	192.168.64.3	TCP	58	35707 → 5900 [S]
38	251.484708919	192.168.64.2	192.168.64.3	TCP	58	35707 → 554 [S]
39	251.484719253	192.168.64.2	192.168.64.3	TCP	58	35707 → 143 [S]
40	251.484730919	192.168.64.2	192.168.64.3	TCP	58	35707 → 587 [S]
41	251.484740253	192.168.64.2	192.168.64.3	TCP	58	35707 → 1025 [S]
42	251.484749627	192.168.64.2	192.168.64.3	TCP	58	35707 → 3389 [S]
43	251.485979993	192.168.64.3	192.168.64.2	TCP	54	993 → 35707 [R]
44	251.485980327	192.168.64.3	192.168.64.2	TCP	54	53 → 35707 [R]
45	251.485980410	192.168.64.3	192.168.64.2	TCP	54	443 → 35707 [R]
46	251.485980452	192.168.64.3	192.168.64.2	TCP	54	995 → 35707 [R]
47	251.485980493	192.168.64.3	192.168.64.2	TCP	54	5900 → 35707 [R]

TCP Traffic Capture in Wireshark



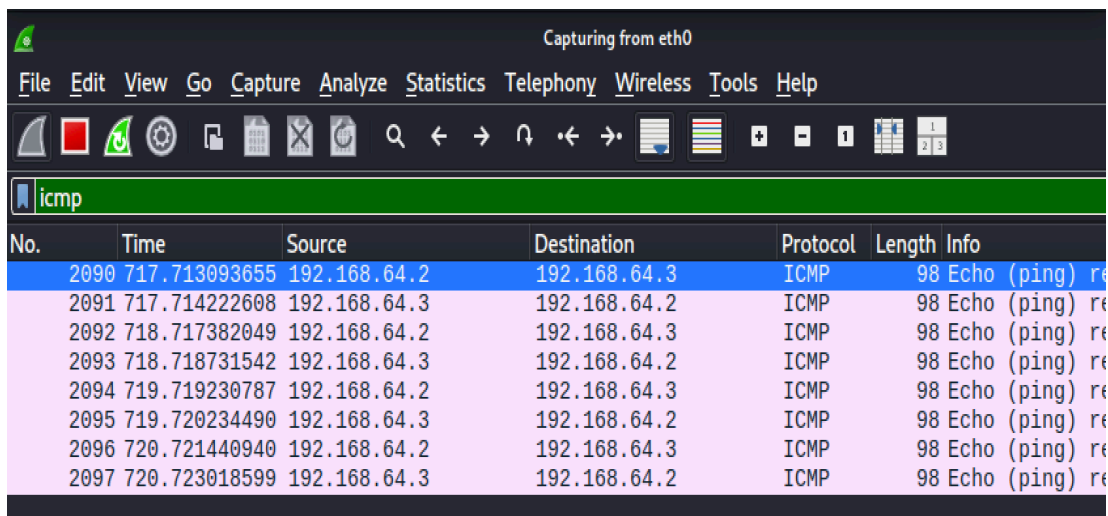
Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp

No.	Time	Source	Destination	Protocol	Length	Info
2	0.200547731	95.216.144.226	192.168.64.2	NTP	90	NTP Version 4...
10	87.121621109	fe80::944e:30ff:fee...	fe80::7c3b:2dff:feb...	DNS	93	Standard query
11	87.121638775	fe80::944e:30ff:fee...	fe80::7c3b:2dff:feb...	DNS	93	Standard query
12	87.229171256	fe80::7c3b:2dff:feb...	fe80::944e:30ff:fee...	DNS	165	Standard query
13	87.229171547	fe80::7c3b:2dff:feb...	fe80::944e:30ff:fee...	DNS	165	Standard query
23	143.213029214	192.168.64.1	192.168.64.255	NBNS	110	Refresh NB MAC
31	250.974979993	fe80::944e:30ff:fee...	fe80::7c3b:2dff:feb...	DNS	105	Standard query
32	250.976463315	fe80::7c3b:2dff:feb...	fe80::944e:30ff:fee...	DNS	164	Standard query
2038	256.249873124	192.168.64.2	95.216.144.226	NTP	90	NTP Version 4...
2039	256.445329368	95.216.144.226	192.168.64.2	NTP	90	NTP Version 4...
2044	306.085811021	192.168.64.1	224.0.0.251	MDNS	262	Standard query
2045	306.085811771	fe80::7c3b:2dff:feb...	ff02::fb	MDNS	282	Standard query
2046	307.087023464	192.168.64.1	224.0.0.251	MDNS	262	Standard query
2047	307.087152172	fe80::7c3b:2dff:feb...	ff02::fb	MDNS	282	Standard query
2048	309.087517127	192.168.64.1	224.0.0.251	MDNS	262	Standard query

UDP Traffic Capture in Wireshark



Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
2090	717.713093655	192.168.64.2	192.168.64.3	ICMP	98	Echo (ping) re
2091	717.714222608	192.168.64.3	192.168.64.2	ICMP	98	Echo (ping) re
2092	718.717382049	192.168.64.2	192.168.64.3	ICMP	98	Echo (ping) re
2093	718.718731542	192.168.64.3	192.168.64.2	ICMP	98	Echo (ping) re
2094	719.719230787	192.168.64.2	192.168.64.3	ICMP	98	Echo (ping) re
2095	719.720234490	192.168.64.3	192.168.64.2	ICMP	98	Echo (ping) re
2096	720.721440940	192.168.64.2	192.168.64.3	ICMP	98	Echo (ping) re
2097	720.723018599	192.168.64.3	192.168.64.2	ICMP	98	Echo (ping) re

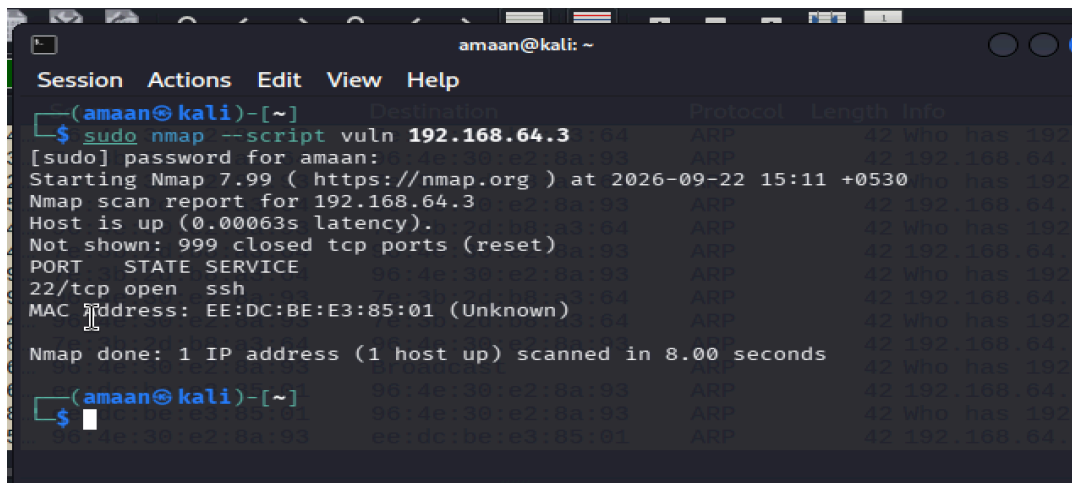
ICMP Traffic Capture in Wireshark

Task 12: Vulnerability Assessment

To identify potential security vulnerabilities and exposed services on the Ubuntu lab system using Nmap NSE vulnerability scripts.

Command used: `sudo nmap --script vuln 192.168.64.3`

The vulnerability assessment identified the Ubuntu system as active with SSH exposed on port 22/tcp. The Nmap vulnerability scripts did not report any specific vulnerabilities for the scanned services. This indicates that no known vulnerability was detected by the selected Nmap scripts during this assessment.



```
Session Actions Edit View Help
(amaan@kali)-[~]
$ sudo nmap --script vuln 192.168.64.3
[sudo] password for amaan:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-22 15:11 +0530
Nmap scan report for 192.168.64.3
Host is up (0.00063s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: EE:DC:BE:E3:85:01 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 8.00 seconds
```

Nmap vulnerability assessment of the Ubuntu lab target

Task 13: System Hardening

Hardening Change 1 Disable Root Login

Root login over SSH was disabled by setting PermitRootLogin no in the SSH configuration. The configuration was validated using sshd -t, and the SSH service was restarted successfully. The service remained active and running after the change.

```
amaan@ubuntu-target:~$ sudo sshd -t
amaan@ubuntu-target:~$ sudo systemctl restart ssh
amaan@ubuntu-target:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-09-22 10:11:22 UTC; 33s ago
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 2624 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 2625 (sshd)
   Tasks: 1 (limit: 4544)
   Memory: 1.1M (peak: 1.5M)
   CPU: 17ms
   CGroup: /system.slice/ssh.service
           └─2625 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 22 10:11:22 ubuntu-target systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 22 10:11:22 ubuntu-target sshd[2625]: Server listening on 0.0.0.0 port 22.
Sep 22 10:11:22 ubuntu-target sshd[2625]: Server listening on :: port 22.
Sep 22 10:11:22 ubuntu-target systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
amaan@ubuntu-target:~$
```

SSH service verified after disabling root login

Hardening Change 2 Enable UFW Firewall

The UFW firewall was enabled on the Ubuntu target system. Incoming connections are denied by default, while outgoing connections are allowed. SSH traffic on port 22 was specifically allowed for authorized remote access. This helps reduce unauthorized network access to the system.

```
amaan@ubuntu-target:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
amaan@ubuntu-target:~$ sudo ufw enable
Firewall is active and enabled on system startup
amaan@ubuntu-target:~$ Command may disrupt existing ssh connections. Proceed with operation (y|n)?
-bash: syntax error near unexpected token `('
amaan@ubuntu-target:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

amaan@ubuntu-target:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)

amaan@ubuntu-target:~$
```

UFW firewall enabled with SSH port 22 allowed

Hardening Change 3: Limit SSH Authentication Attempts

The SSH configuration was hardened by reducing MaxAuthTries from 6 to 3. This limits the number of unsuccessful authentication attempts allowed for an SSH connection and helps reduce password-guessing attempts. The configuration was validated using `sshd -t`, and the SSH service was restarted successfully.

```
amaan@ubuntu-target:~$ sudo sshd -t
amaan@ubuntu-target:~$ sudo systemctl restart ssh
amaan@ubuntu-target:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-09-22 10:42:10 UTC; 21s ago
 TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 2892 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 2893 (sshd)
    Tasks: 1 (limit: 4544)
   Memory: 1.1M (peak: 1.5M)
      CPU: 16ms
   CGroup: /system.slice/ssh.service
           └─2893 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 22 10:42:10 ubuntu-target systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 22 10:42:10 ubuntu-target sshd[2893]: Server listening on 0.0.0.0 port 22.
Sep 22 10:42:10 ubuntu-target sshd[2893]: Server listening on :: port 22.
Sep 22 10:42:10 ubuntu-target systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
amaan@ubuntu-target:~$
```

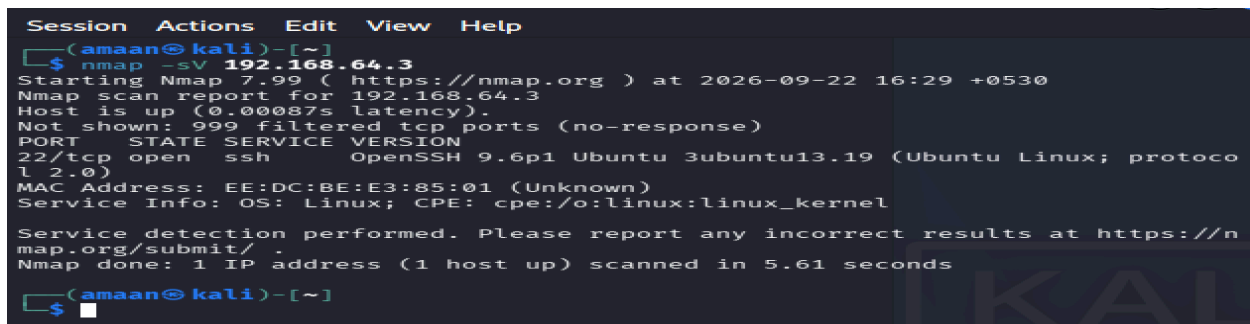
SSH service verified after limiting authentication attempts to 3

Task 14: Final Mini Security Assessment

The final assessment was performed to verify the security status of the Ubuntu target after applying the hardening changes.

Assessment Steps:

1. Verified that the Ubuntu target was reachable from Kali Linux.
2. Performed an Nmap service scan to identify open ports and services.
3. Verified that SSH was running on port 22.
4. Verified that the UFW firewall was active.
5. Verified that root SSH login was disabled.
6. Verified that SSH authentication attempts were limited to 3.



```
Session Actions Edit View Help
(amaaan@kali)-[~]
$ nmap -sV 192.168.64.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-22 16:29 +0530
Nmap scan report for 192.168.64.3
Host is up (0.00087s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.19 (Ubuntu Linux; protocol 2.0)
MAC Address: EE:DC:BE:E3:85:01 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.61 seconds
(amaaan@kali)-[~]
$
```

Final Nmap service scan of the hardened Ubuntu target

The final Nmap scan confirmed that the Ubuntu target at 192.168.64.3 was reachable and that SSH was available on port 22. The assessment also confirmed that the system hardening changes were applied, including disabling root SSH login, enabling the UFW firewall, and limiting SSH authentication attempts to 3.

Conclusion: This practical demonstrated the use of Nmap and Wireshark for network discovery, service identification, packet analysis, vulnerability assessment, and system hardening. The Ubuntu target was assessed in an authorized lab environment, and security improvements were applied through SSH configuration and UFW firewall protection.

Learning Outcomes

- Learned network discovery using Nmap.
- Learned to identify open ports and running services.
- Learned to capture and analyze network packets using Wireshark.
- Learned to analyze ARP, DNS, TCP, UDP, and ICMP traffic.
- Learned basic vulnerability assessment techniques.
- Learned basic Linux system-hardening techniques.
- Learned how firewall and SSH configuration can improve system security.